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Звіт до лабораторної роботи 2

Performance Measurement and Analysis.

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1.Середовище виконання лабораторної роботи

```
llucc@ASUSPC:~/lab-2-sp$ g++ --version
g++ (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

llucc@ASUSPC:~/lab-2-sp$ /usr/bin/time --version
time (GNU Time) UNKNOWN
Copyright (C) 2018 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Written by David Keppel, David MacKenzie, and Assaf Gordon.
llucc@ASUSPC:~/lab-2-sp$ perf --version
perf version 7.0.14
llucc@ASUSPC:~/lab-2-sp$ uname -r
6.6.87.2-microsoft-standard-WSL2
llucc@ASUSPC:~/lab-2-sp$
```

2.Створення досліджуваної програми

```
llucc@ASUSPC:~/lab-2-sp$ cat > program.cpp <<'EOF'
#include <iostream>
#include <vector>
#include <cstddef>

int main() {
    constexpr std::size_t N = 1048;

    std::vector<double> a(N * N, 1.0);
    std::vector<double> b(N * N, 2.0);
    std::vector<double> c(N * N, 0.0);

    for (std::size_t i = 0; i < N; ++i) {
        for (std::size_t j = 0; j < N; ++j) {
            for (std::size_t k = 0; k < N; ++k) {
                c[i * N + j] +=
                    a[i * N + k] * b[k * N + j];
            }
        }
    }

    std::cout << "Result: " << c[0] << '\n';
    return 0;
}
EOF
```

```
llucc@ASUSPC:~/lab-2-sp$ g++ -O0 -g -fno-omit-frame-pointer program.cpp -o program_O0
llucc@ASUSPC:~/lab-2-sp$ g++ -O2 -g -fno-omit-frame-pointer program.cpp -o program_O2
llucc@ASUSPC:~/lab-2-sp$ ./program_O0
Result: 2096
llucc@ASUSPC:~/lab-2-sp$ ./program_O2
Result: 2096
```

3.Вимірювання часу та пам'яті

```
llucc@ASUSPC:~/lab-2-sp$ cat time_O0.txt
        Command being timed: "./program_O0"
        User time (seconds): 10.75
        System time (seconds): 0.03
        Percent of CPU this job got: 99%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:10.81
        Average shared text size (kbytes): 0
        Average unshared data size (kbytes): 0
        Average stack size (kbytes): 0
        Average total size (kbytes): 0
        Maximum resident set size (kbytes): 29568
        Average resident set size (kbytes): 0
        Major (requiring I/O) page faults: 11
        Minor (reclaiming a frame) page faults: 6583
        Voluntary context switches: 10
        Involuntary context switches: 14
        Swaps: 0
        File system inputs: 1104
        File system outputs: 0
        Socket messages sent: 0
        Socket messages received: 0
        Signals delivered: 0
        Page size (bytes): 4096
        Exit status: 0
```

Заміри для не оптимізованої програми

```
llucc@ASUSPC:~/lab-2-sp$ cat time_O2.txt
        Command being timed: "./program_O2"
        User time (seconds): 0.41
        System time (seconds): 0.00
        Percent of CPU this job got: 98%
        Elapsed (wall clock) time (h:mm:ss or m:ss): 0:00.43
        Average shared text size (kbytes): 0
        Average unshared data size (kbytes): 0
        Average stack size (kbytes): 0
        Average total size (kbytes): 0
        Maximum resident set size (kbytes): 29568
        Average resident set size (kbytes): 0
        Major (requiring I/O) page faults: 1
        Minor (reclaiming a frame) page faults: 6592
        Voluntary context switches: 1
        Involuntary context switches: 0
        Swaps: 0
        File system inputs: 96
        File system outputs: 0
        Socket messages sent: 0
        Socket messages received: 0
        Signals delivered: 0
        Page size (bytes): 4096
        Exit status: 0
```

Заміри для оптимізованої програми

4.Профілювання за допомогою perf

```
Result: 2096

 Performance counter stats for './program_O2':

            430.47 msec task-clock:u
                 0      context-switches:u
                 0      cpu-migrations:u
              6576      page-faults:u

       0.433801943 seconds time elapsed

       0.418948000 seconds user
       0.012085000 seconds sys


llucc@ASUSPC:~/lab-2-sp$ perf stat -e task-clock,context-switches,cpu-migrations,page-faults -- ./program_O0
Result: 2096

 Performance counter stats for './program_O0':

           9991.60 msec task-clock:u
                 0      context-switches:u
                 0      cpu-migrations:u
              6576      page-faults:u

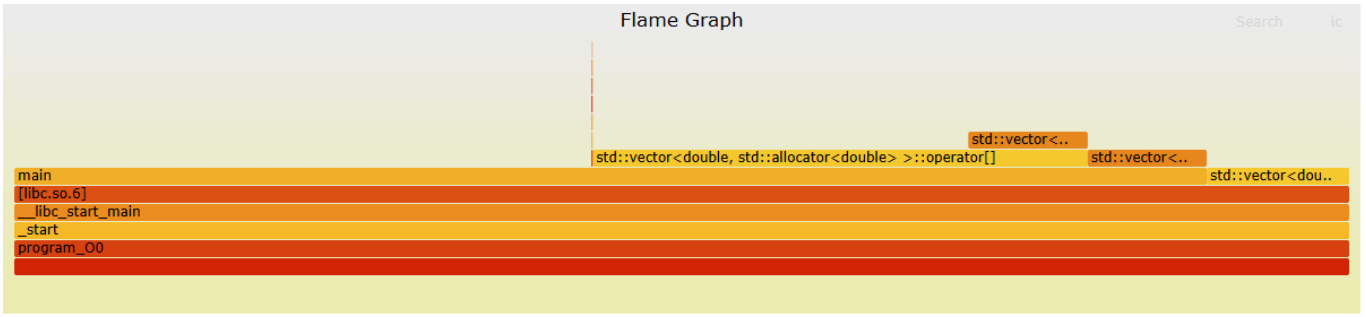
      10.007376887 seconds time elapsed

       9.980507000 seconds user
       0.012005000 seconds sys
```

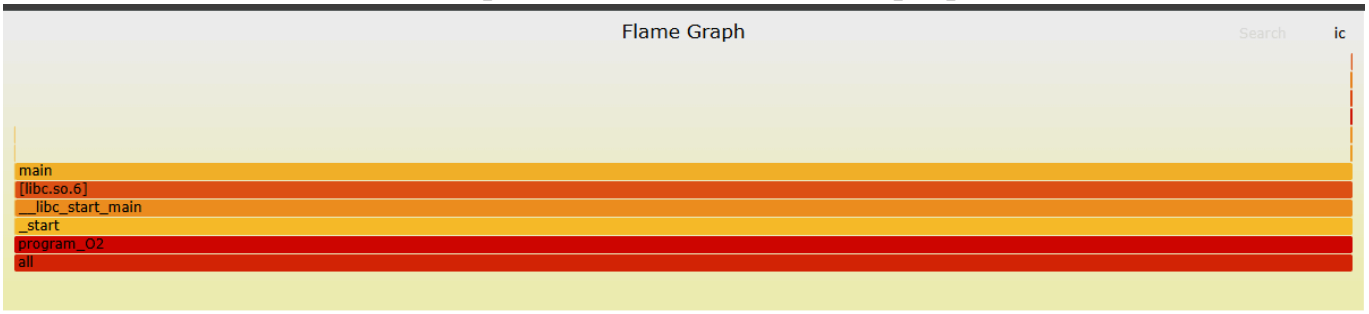
```
Samples: 36K of event 'cpu-clock:u', Event count (approx.): 9202000000
  Children      Self  Command     Shared Object         Symbol
+  100.00%     0.00%  program_O0  program_O0            [.] _start
+  100.00%     0.00%  program_O0  libc.so.6            [.] __libc_start_main
+  100.00%     0.00%  program_O0  libc.so.6            [.] 0x00007ddc55a2a601
+   89.33%    43.24%  program_O0  program_O0            [.] main
+   47.76%    38.81%  program_O0  program_O0            [.] std::vector<double, std::allocator<double> >::operator[](uns
+   17.86%    17.86%  program_O0  program_O0            [.] std::vector<double, std::allocator<double> >::size() const
     0.08%     0.00%  program_O0  program_O0            [.] std::allocator<double>::~allocator() (inlined)
     0.08%     0.00%  program_O0  program_O0            [.] std::vector<double, std::allocator<double> >::vector(unsigne
     0.08%     0.00%  program_O0  program_O0            [.] std::vector<double, std::allocator<double> >::_M_fill_initia
     0.08%     0.00%  program_O0  program_O0            [.] double* std::__uninitialized_fill_n_a<double*, unsigned long
     0.08%     0.00%  program_O0  program_O0            [.] double* std::uninitialized_fill_n<double*, unsigned long, do
     0.07%     0.03%  program_O0  program_O0            [.] double* std::__do_uninit_fill_n<double*, unsigned long, doub
     0.05%     0.05%  program_O0  program_O0            [.] void std::_Construct<double, double const&>(double*, double
     0.01%     0.01%  program_O0  program_O0            [.] operator new(unsigned long, void*)
     0.01%     0.00%  program_O0  program_O0            [.] double* std::__addressof<double>(double&) (inlined)
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x000000000000c6e3
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560f7348
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560f84e1
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560f6dc6
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560fa5c3
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560e8581
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560e5051
     0.00%     0.00%  program_O0  ld-linux-x86-64.so.2  [.] 0x00007ddc560e06e3
```

```
Samples: 1K of event 'cpu-clock:u', Event count (approx.): 445500000
  Children      Self  Command     Shared Object         Symbol
+   99.94%    99.94%  program_O2  program_O2            [.] main
+   99.94%     0.00%  program_O2  program_O2            [.] _start
+   99.94%     0.00%  program_O2  libc.so.6            [.] __libc_start_main
+   99.94%     0.00%  program_O2  libc.so.6            [.] 0x00007629a942a601
     0.17%     0.00%  program_O2  program_O2            [.] std::vector<double, std::allocator<double> >::vector(unsigne
     0.17%     0.00%  program_O2  program_O2            [.] std::vector<double, std::allocator<double> >::_M_fill_initia
     0.17%     0.00%  program_O2  program_O2            [.] double* std::__uninitialized_fill_n_a<double*, unsigned long
     0.17%     0.00%  program_O2  program_O2            [.] double* std::uninitialized_fill_n<double*, unsigned long, do
     0.17%     0.00%  program_O2  program_O2            [.] double* std::__do_uninit_fill_n<double*, unsigned long, doub
     0.11%     0.00%  program_O2  program_O2            [.] void std::_Construct<double, double const&>(double*, double
     0.06%     0.06%  program_O2  ld-linux-x86-64.so.2  [.] 0x0000000000010f92
     0.06%     0.00%  program_O2  ld-linux-x86-64.so.2  [.] 0x00007629a9cad348
     0.06%     0.00%  program_O2  ld-linux-x86-64.so.2  [.] 0x00007629a9cae4e1
     0.06%     0.00%  program_O2  ld-linux-x86-64.so.2  [.] 0x00007629a9cacdc6
     0.06%     0.00%  program_O2  ld-linux-x86-64.so.2  [.] 0x00007629a9cb05c3
     0.06%     0.00%  program_O2  ld-linux-x86-64.so.2  [.] 0x00007629a9c9e581
     0.06%     0.00%  program_O2  ld-linux-x86-64.so.2  [.] 0x00007629a9c9af92
```

5.FlameGraph



FlameGraph для не оптимізованої програми



FlameGraph для оптимізованої програми

6. Енергоспоживання

У стані спокою споживання від 2.5 до 7.6 Вт, в середньому це 5.2
При виконанні програми -O0 споживання в середньому було 25.5 Вт
Тоді сама програма споживала (25.5 - 5.2) = 20.3 Вт
При виконанні програми -O2 споживання в середньому було 10.8 Вт
Тоді сама програма споживала (10.8 - 5.2) = 5.6 Вт

Результати у Додатку 1

7.Порівняння

Показник	-O0	-O2
Час виконання	10.75	0.42
Максимальна пам'ять	29568	29568
CPU task-clock	100.00%	99.94%
Кількість перемикань контексту	0	0
Енергоспоживання	10.8	5.6

```
C:\Users\llucc>typeperf "\Energy Meter(*)\Power" -si 1

"(PDH-CSV 4.0)","\\ASUS_PC\Energy Meter(RAPL_Package0_DRAM)\Power","\\ASUS_PC\Energy Meter(RAPL_Pac
\Power","\\ASUS_PC\Energy Meter(_Total)\Power"
"10/24/2026 12:34:08.478","0.000000","6109.595437","2899.464791","19.685694","0.000000"
"10/24/2026 12:34:09.486","0.000000","6814.146823","3617.008808","31.325517","0.000000"
"10/24/2026 12:34:10.492","0.000000","8267.445808","4645.559228","28.941733","0.000000"
"10/24/2026 12:34:11.505","0.000000","5253.919346","2603.836532","18.996080","0.000000"
"10/24/2026 12:34:12.512","0.000000","2249.724490","418.090929","9.821742","0.000000"
"10/24/2026 12:34:13.514","0.000000","2439.854385","487.863053","7.000780","0.000000"
"10/24/2026 12:34:14.525","0.000000","3334.476302","1178.432165","7.183354","0.000000"
"10/24/2026 12:34:15.530","0.000000","7696.501432","5108.743937","9.460752","0.000000"
"10/24/2026 12:34:16.544","0.000000","26459.630440","22539.040714","6.621252","0.000000"
"10/24/2026 12:34:17.544","0.000000","25856.396321","22004.232813","6.892696","0.000000"
"10/24/2026 12:34:18.556","0.000000","25404.337974","21424.777314","6.932980","0.000000"
"10/24/2026 12:34:19.574","0.000000","26169.390842","22350.413521","6.292702","0.000000"
"10/24/2026 12:34:20.587","0.000000","23685.159129","19774.388649","6.983922","0.000000"
"10/24/2026 12:34:21.589","0.000000","25973.265800","22066.271124","6.944546","0.000000"
"10/24/2026 12:34:22.598","0.000000","24835.175678","21059.486694","7.190116","0.000000"
"10/24/2026 12:34:23.613","0.000000","25541.185665","21599.807601","9.018757","0.000000"
"10/24/2026 12:34:24.620","0.000000","2292.328346","359.596255","6.237524","0.000000"
"10/24/2026 12:34:25.620","0.000000","2275.429910","340.011495","6.825828","0.000000"
"10/24/2026 12:34:26.635","0.000000","2131.470434","347.688417","6.192618","0.000000"
```

```
llucc@ASUSPC:~/lab-2-sp$ ^C
llucc@ASUSPC:~/lab-2-sp$ perf stat -e task-clock,context-switches,cpu-mig
Result: 2096

 Performance counter stats for './program_O0':

          8229.02 msec task-clock:u
                0      context-switches:u
                0      cpu-migrations:u
             6575      page-faults:u

       8.235047951 seconds time elapsed

       8.217758000 seconds user
       0.012008000 seconds sys


llucc@ASUSPC:~/lab-2-sp$
```

```
C:\Users\llucc>typeperf "\Energy Meter(*)\Power" -si 1

"(PDH-CSV 4.0)","\\ASUS_PC\Energy Meter(RAPL_Package0_DRAM)\Power","\\ASUS_PC\Energy Meter
\Power","\\ASUS_PC\Energy Meter(_Total)\Power"
"10/24/2026 12:35:56.864","0.000000","6788.090507","3342.963501","27.829921","0.000000"
"10/24/2026 12:35:57.870","0.000000","7367.977326","3835.654501","23.701488","0.000000"
"10/24/2026 12:35:58.875","0.000000","8876.818175","5303.535075","37.471723","0.000000"
"10/24/2026 12:35:59.877","0.000000","5024.679206","2496.939545","11.747866","0.000000"
"10/24/2026 12:36:00.879","0.000000","10816.274117","7970.913233","8.632256","0.000000"
"10/24/2026 12:36:01.894","0.000000","2531.325715","315.604777","6.307288","0.000000"
"10/24/2026 12:36:02.901","0.000000","3660.534003","1359.008855","6.547687","0.000000"
"10/24/2026 12:36:03.915","0.000000","3153.528810","855.805568","16.558539","0.000000"

The command completed successfully.
```

```
llucc@ASUSPC:~/lab-2-sp$ perf stat -e task-clock,context-switches,cpu-migrations,page-faults
Result: 2096

 Performance counter stats for './program_O2':

           425.85 msec task-clock:u
                0      context-switches:u
                0      cpu-migrations:u
             6577      page-faults:u

       0.426312958 seconds time elapsed

       0.410238000 seconds user
       0.015931000 seconds sys
```